

Production Supporting Systems in Factories

ระบบสนับสนุนการผลิตในโรงงานอุตสาหกรรม

Cloud Database

InfluxDB

- InfluxDB is a time series database (TSDB) developed by the company InfluxData.
- It is used for storage and retrieval of time series data in fields such as operations monitoring, application metrics, Internet of Things sensor data, and real-time analytics.
- Ranking
- Performance

Get Started with InfluxDB

- Register for an account at [InfluxDB Cloud](#).
- Create a new InfluxDB instance.
- Create a new organization and a new bucket.
- Create a new API token with all access permissions.

You will need

```
CLUSTER_URL=  
ORGANIZATION_NAME=  
BUCKET_NAME=  
API_TOKEN=
```

InfluxDB Data Model

- **Organization:** A container for all your InfluxDB resources.
- **Bucket:** A named location where time series data is stored.
- **Measurement:** A container for time series data, similar to a table in relational databases.
- **Tag:** A key-value pair that is used to store metadata about a measurement.
- **Field:** A key-value pair that stores the actual data points in a measurement.

Query Languages

- InfluxDB supports two query languages: `InfluxQL` and `Flux`.
- `InfluxQL` is similar to `SQL` and is used for querying time series data.
 - You will see this in Data Explorer in InfluxDB Cloud.
- `Flux` is a more powerful and flexible query language that allows for complex data processing and analysis.
 - You will use this in Node-RED to query data from InfluxDB.

Example of InfluxQL Query:

```
SELECT mean("value") FROM "your_measurement_name" WHERE time > now() - 1h GROUP BY time(1m)
```

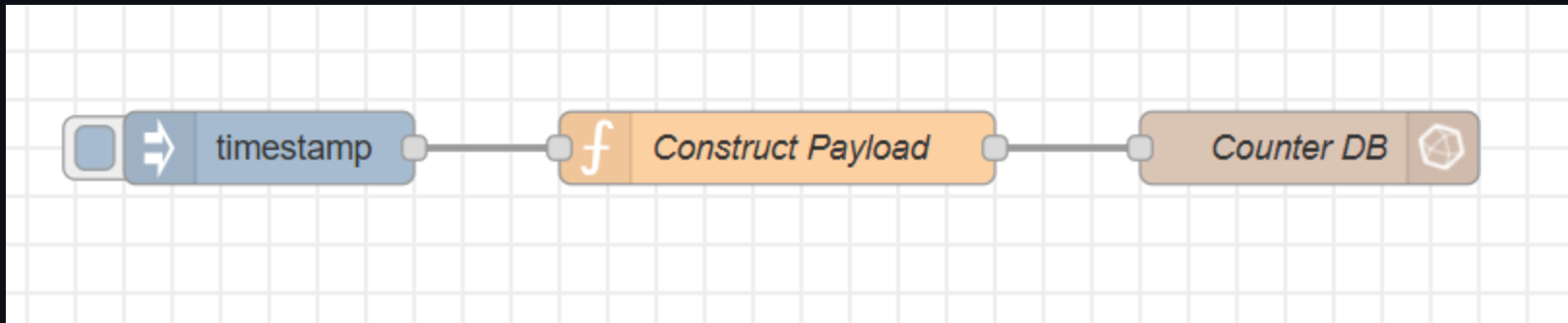
Example of Flux Query:

```
from(bucket: "your_bucket_name")
  |> range(start: -1h)
  |> filter(fn: (r) => r._measurement == "your_measurement_name")
  |> group(columns: ["_field"])
  |> mean()
```

InfluxDB Node-RED Integration

- Use the `node-red-contrib-influxdb` package to integrate InfluxDB with Node-RED.

Write Data to InfluxDB



InfluxDB Node

Edit influxdb out node

Delete Cancel Done

Properties

Name Counter DB

Server [v2.0] https://us-east-1-1.aws.clo [edit] [add]

Organization Dev Team

Bucket my_bucket

Measurement counter

Time Precision Milliseconds (ms)

InfluxDB Configuration Node

Edit influxdb out node > **Edit influxdb node**

Delete Cancel **Update**

Properties

Name

Version

URL

Token

Connection timeout (seconds)

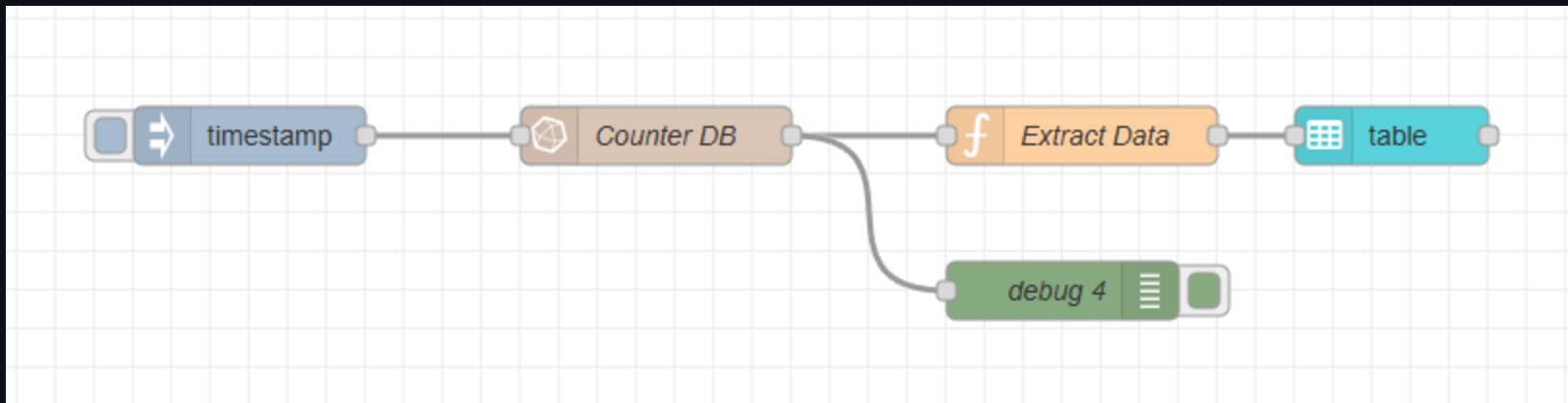
☒ Verify server certificate

- Choose InfluxDB v2, not v1.
- Use the API token for authentication.

Construct Payload Function Node

```
const field = {  
  counter: msg.payload,  
};  
const tag = {  
  tag_1: "TAG_1",  
  tag_2: "TAG_2",  
};  
msg.payload = [field, tag];  
return msg;
```

Receive Data from InfluxDB



Flux Query to Get Last 5 Records

```
from(bucket: "my_bucket") // replace with your bucket name
  |> range(start: -2h)
  |> filter(fn: (r) => r._measurement == "counter" and r._field == "counter") // replace with your measurement and field
  |> group() // remove grouping so you get a single table
  |> sort(columns: ["_time"], desc: true)
  |> limit(n: 5)
```

Extract Data Function Node

```
const _payload = msg.payload;

msg.payload = _payload.map((d) => {
  const _dt = new Date(d["_time"]);
  const dt = _dt.toLocaleString(); // e.g. "11/30/2025, 9:37:46 AM"
  return {
    "Date/Time": dt,
    Tag_1: d["tag_1"], // replace with your actual tag key
    Tag_2: d["tag_2"], // replace with your actual tag key
    Count: d["_value"],
  };
});
return msg;
```

Table Output

Edit table node

Delete Cancel Done

Properties

Name Name

Group [Page 1] Group 1

Size auto

Label optional label

Class Optional CSS class name(s)

Max Rows 0

Action Replace

Breakpoint defaults: Mobile (< 576px)